Lecture 1 – Introduction

Financial Markets and Intermediaries

Financial Markets:

- Markets in which funds from people and firms who have an excess of available funds are transferred to people and firms who have a need of funds

Financial Intermediaries:

- Institutions that borrow funds from agents who have saved and make funds available to other agents in the economy
- Their advantage is that they can gather information and reduce risk

Functions of Financial Markets and Financial Intermediaries:

- Allocational efficiency: Promotes economic efficiency by producing an efficient allocation of capital, which increases production
- Consumption smoothing: Directly improves the well-being of consumers by allowing them to time purchases better

3 Key Services of Financial Intermediaries

1: Brokerage

- Brokers bring together transactors in financial claims with complementary needs
- Brokerage overcomes information asymmetries between (potential) transactors in financial claims (e.g. lenders and borrowers):
    - Pre-contractual: Adverse selection Duplicated Screening
    - Post-contractual: Moral hazard
- Helps with information reusability:
    - **Cross-sectional information reusability**: Screening/monitoring of one firm/individual helps screening/monitoring other firms/individuals
    - **Intertemporal information reusability**: Having interacted with a firm/individual in the past helps screening/monitoring of this firm/individual in the future

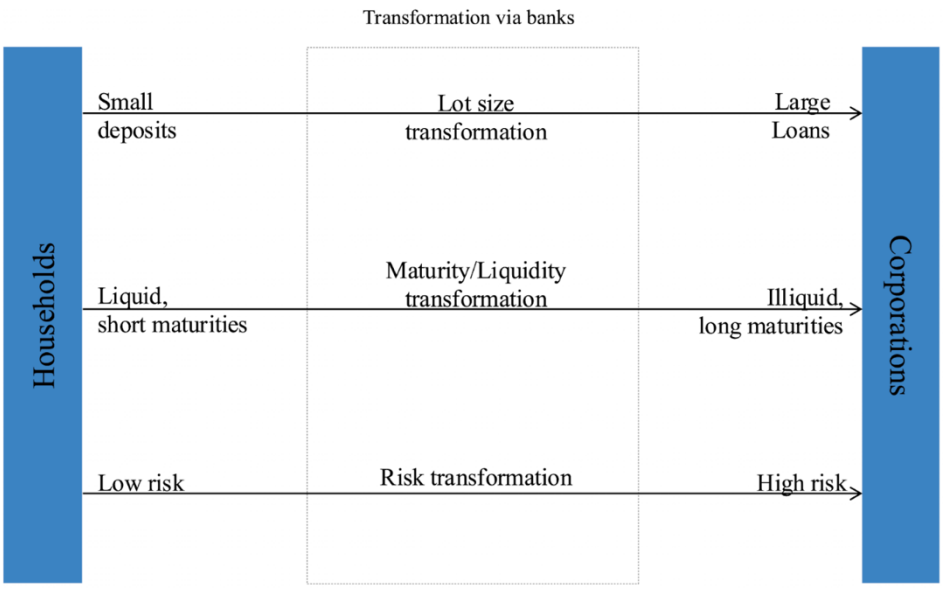
2: Qualitative Asset Transformation

- QAT overcomes structural mismatches in the claims that financial transactors want to hold

3: Transaction Services ("Infrastructure")

- Banks provide the financial infrastructure for payment and custody services
    - 1. Cash services
    - 2. Transfer + Cards
    - 3. Securities services
- Banks only facilitate matching and do NOT act as principal in the purchase (sale)

- Three main transformation functions of banks:
    - 1. Lot size transformation
    - 2. Maturity/Liquidity transformation
    - 3. Risk transformation



Lecture 2 – Bank balance sheets and P&L statements

Types of Banks

- Three typical types of bank business models

<u>1.) Savings banks</u>

- deposits and lending
- income from net interest margin
- e.g. Sparkasse (traditional bank)

<u>2.) Broker/Asset Manager</u>

- small and "boring" balance sheet
  - no risk on balance sheet
- large AuM
- main income from fees
- e.g. Trade Republic

<u>3.) Universal bank</u>

- combining both models
- adding other areas such as trading business
- commercial bank with complex activities
- e.g. UBS

Balance Sheet Ratios

<u>Equity Ratio</u>

$$Equity\ Ratio = \frac{Equity}{Total\ Assets}$$

- ability to absorb losses
- Typical ratio for a bank is low
- Non-financial corporations have a 50% equity ratio on average

<u>Loan-to-Deposit-Ratio (LDR)</u>

$$LDR = \frac{Loans}{Deposits}$$

- Demand/supply for loans and deposits
- Ratio below 100% is typical because banks need to hold liquidity (cash/securities) to cover potential outflows of deposits

<u>Proportion of short-term financing</u>

$$Proportion\ of\ shortterm\ financing = \frac{Overnight\ financing}{Total\ liabilities}$$

- Fragility with respect to changes in funding conditions

<u>Liquid-Assets-Ratio</u>

$$Liquid\ Assets\ Ratio = \frac{Cash + Securities}{Total\ Assets}$$

- Ability to absorb funding shocks
- Bank could withstand bank run on X% of liabilities

P&L Ratios

<u>Average interest rate earned from cash, securities, and loans</u>

$$R_{CLS} = \frac{Interest\ Income}{Cash + Securities + Loans}$$

- depends on: interest level in economy; composition cash vs securities vs loans (higher liquidity implies lower rates); riskiness of loans and securities; operational costs of originating loans
- asset composition: cash → a very low return, loans → higher return (but less liquid)

<u>Average interest expense on deposits + other liabilities</u>

$$R_{dep} = \frac{Interest\ Expense}{Deposits}$$

- Deposits usually with low interest rates, but service component (free or cheap transaction account)

<u>Interest Margin</u>

$$Interest\ Margin = R_{CLS} - R_{dep}$$

- drivers: Competition, Credit risk, Liquidity and maturity transformation

<u>Cost-to-Income-Ratio</u>

$$Cost\ to\ Income\ Ratio = \frac{Salary\ and\ administrative\ expenses}{Net\ Revenues}$$

$$Net\ Revenues = Net\ interest\ income + Fee\ and\ commission\ income + Trading\ income$$

- Typical measure for the efficiency of banks
- Note: Inclusion/exclusion of other income/costs and loan loss provisions varies from case to case

<u>Return on Equity (after Taxes)</u>

$$Return\ on\ Equity\ (after\ Taxes) = \frac{Income\ after\ Taxes}{Equity}$$

- Profitability measure

Lecture 3 – Bank lending 1

Interbanks

- Banks mostly lend to each other (short term or overnight) to manage liquidity
- Interbank market is quite efficient but it causes a chain of collapses in case one bank collapses because the banks are connected

Loans

- Definition: Purchase of an illiquid and customized financial asset
    - Asset: claim against a borrower's future cash flow
    - Price: loan is the payment to obtain the asset

Types of loans

- Term loan
    - Typically used to finance fixed investments
    - Paid out in full on day 1. Paid back with bullet structure or amortization schedule
- Revolver (=line of credit)
    - Borrower can borrow and repay and re-borrow money during the availability period – it "revolves"
    - Similar to „overdrafts" or credit cards for private customers
- Letter of credit
    - Similar to a guarantee, used primarily for exports/imports

Financial obligations of a Loan Contract

- Pricing (interest, fees)
- Repayment schedule
- Take-down schedule (for credit lines)

Non-financial obligations of a Loan Contract

- **Warranties**: set conditions at time of contracting.
    - Examples: financial statements are real, borrower has a valid title of all assets, borrower is not involved in litigation.
- **Covenants**: restrict borrower actions during the contract
    - Examples for <u>affirmative covenants</u>: provision of periodical information, maintanance of insurance, etc.
    - Examples of <u>negative covenants</u>:
        - *restrictive covenants*: prohibition of asset sales or mergers, restrictions on dividends, etc.
        - *financial covenants*: limit to the debt-to-EBITDA ratio, etc.
    - Default provisions: conditions under which loan becomes callable

Rationale for Covenants

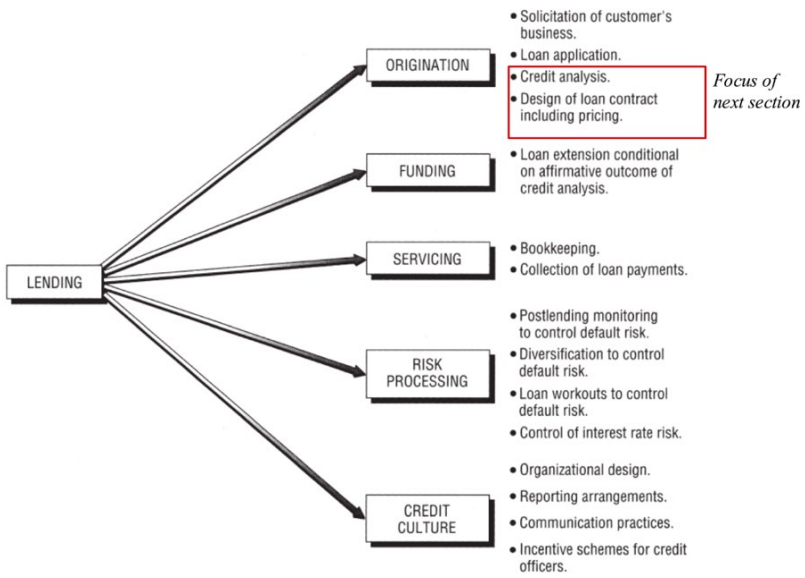
- Similar to Performance Pricing (Lecture 5)
- Covenants used for large changes (avoids death spiral via performance pricing)

Loans vs. Securities

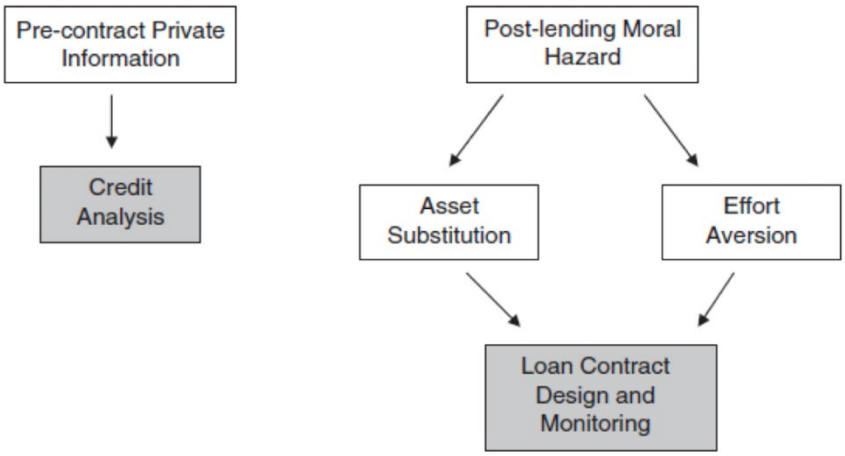
- Securities are usually more liquid
- Securities are traded in secondary markets

- Information asymmetry regarding value is stronger for loans
- Erosion of the distinction between loans and securities
    - Loan sales
    - Securitization

The Lending Process



Information Problems in Loan Contracting

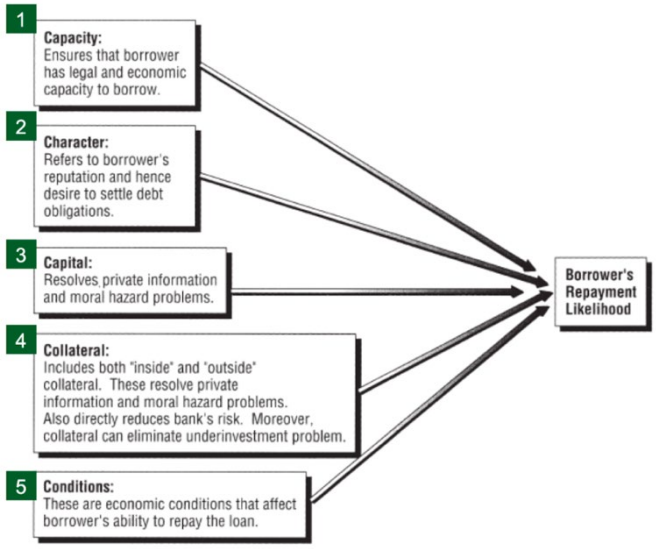


Credit analysis and loan contract design

Determination of the creditworthiness of the borrower:

- 1. Ability to repay the loan
    - The ability of a borrower to repay depends on his/her economic prospects, but also on the broader economic conditions
- 2. Willingness to repay the loan
    - The willingness of a borrower to repay depends on character and the stakes of the borrower in the investment (business / home) getting an unwilling borrower to repay a loan is (very) costly
- Contract design provides incentives for the borrower to behave in the best interest of debtors after a loan has been granted
- Credit analysis and loan contract design are tools to overcome information asymmetries, i.e. to reduce adverse selection and moral hazard

The 5 C's - Credit analysis and loan contract design



-> Factors considered in Credit Analysis

Capacity

- Legal capacity: Is person requesting the loan legally capable of borrowing?
- Economic capacity: Evaluate borrowers' future cash flows available to service debt
    - Economic capacity in corporate lending: financial statement analysis
        - liquidity ratios
        - turnover/activity ratios
        - profitability ratios
    - Economic capacity in mortgage/consumer lending
        - income/expenditure analysis
        - leverage

Character

- borrowers reputation for meeting payment obligations
- external sources: credit bureau
- internal source: bank records

Capital & **C**ollateral

<u>Functions</u>:

- Reduce credit loss directly
    - equity stake/collateral reduces the expected loss in case of default for the lender
- Mitigate pre-contractual private information
    - good borrowers have less to lose by providing equity/collateral than bad ones → Equity/collateral is a credible signal of low risk
- Mitigate post-contractual moral hazard
    - borrowers with an equity stake or collateral have more to lose if the project fails → Equity/collateral provides incentives to choose less risky projects, to provide more effort, and to repay loans

(Economic) **C**onditions

- corporate lending: Company & industry outlook
    - competition
    - customer relationships
    - supply risks
    - pending legislation in industry
- household lending
    - job security
    - real estate market

Lecture 4 – Bank Lending 2

Loan Pricing Equation

Condition for positive expected return on a loan for the bank:

$$\text{Loan Interest Rate} \geq \left[\frac{\text{Institutional Costs}}{\text{Loan Size}}\right] \quad (1)$$

$$+\left[\frac{\text{Expected Loss on Loan}}{\text{Loan Size}}\right] \quad (2)$$

$$+\left[\frac{\text{Debt Financing in Loan}}{\text{Loan Size}} \times \text{Bank's Cost of Debt}\right] \quad (3)$$

$$+\left[\frac{\text{Equity Financing in Loan}}{\text{Loan Size}} \times \text{Bank's Cost of Equity}\right] \quad (4)$$

- As loan size increases, interest rates tend to decrease
- Interest rates should vary with the institutional costs, credit risk and funding costs

<u>Institutional Costs</u>

- Direct costs of screening the applicant
- Direct costs of monitoring the loan and the collateral
- Allocated overhead costs
    - Costs of using property, plant and equipment
    - Costs of management

<u>Expected Loss (EL)</u>

$EL = EaD \times PD \times LGD$

EL: Bank's expected loss on a loan
EaD: Exposure at Default
PD: Probability of Default
LGD: Expected loss given Default

<u>Debt and Equity Financing</u>

- Corporate finance: firms with more volatile cash flows and higher asset betas (greater systematic risk) use more equity in their capital structure
- Basel III!

Credit Rationing

- Credit rationing arises when a borrower does not receive the loan amount he requests at the terms the bank has offered him

<u>Types of Credit Rationing</u>

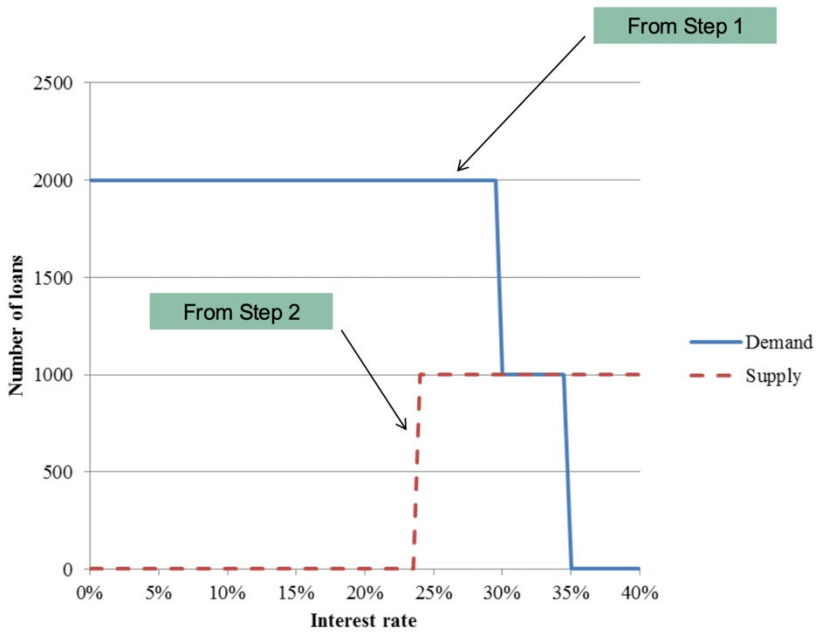
- Type 1 Rationing: The outright refusal of a credit
- Type 2 Rationing: Applicants get less than the requested amount
- Empirical predictions:
    - Rationing should be stronger for borrowers who are less transparent
    - Rationing should be stronger at first interaction of bank and borrower

Expected payoff to borrowers from a loan at interest rate r

$E[\pi] = p \times (R + 1) \times I$

- P = Expected payback probability
- Applicants only borrow if expected Payoff > 0

Loan Demand and Loan Supply



Relationship Banking

- A lender-borrower relationship in which the bank acquires private information about the borrower
- through repeated lending (repeated loans)
- through cross-selling (e.g. current account and loan)

Why is Relationship Banking so prevalent?

- Mitigates precontractual private information
  - The bank learns about the borrower (character) and his economic activity (capacity, conditions)
- Mitigates post-lending moral hazard
  - repeated interaction provides dynamic incentives for the borrower to behave in line with the bank's interests
  - not to choose excessively risky projects
  - provide high effort

Migitating Adverse Selecting in Lending

- Two types of information are important to mitigate adverse selection in lending decisions:
  - information about the borrower
  - information about whether other banks know more or less about the borrower

Lecture 5 – Investment Banking

Syndicated Lending

Syndicated Loan

- A credit granted by a group of lenders to one borrower
- *Lead arranger* negotiates the terms and conditions of the loan with the prospective borrower and participants provide a share of the funding
    - Each syndicate member has a separate claim on the borrower but there is only 1 loan agreement

Main forms of debt

1. Bank loans
2. Bonds
3. Syndicated loans

- Tradeoff between liquidity and flexibility
    - Bank loans = flexible but illiquid
    - Bonds = liquid but inflexible
    - Syndicated loans = something inbetween

Performance Pricing

- The interest rate on a loan depends on the borrower's credit quality over time
    - Migitates post-contractual moral hazard
    - (Same idea as incentive pay: better performance -> lower cost)
- Used in ca. 50% of loans

Rationale for Performance Pricing

- Provide incentives for firms to remain financially sound
    - lower profits, higher leverage or other deterioration in financial conditions leads to higher interest payments
- Screen borrowers as to whether they are financially sound
    - borrowers who know that their creditworthiness will decline are less likely to agree to performance pricing
- Exploit behavioral biases of CEOs/CFOs
    - most CEOs/CFOs are overoptimistic, thinking erroneously that their creditworthiness will very likely improve

Limits:

- a very large increase in interest rates could bring firms into financial difficulties
    - „death spiral“: decrease in creditworthiness (e.g., rating) increases funding costs, increase in funding costs decreases creditworthiness further which leads to another increase in funding costs with performance-based pricing, …

Rationale for Syndicated Lending

Borrower's View

- Efficiency argument:
    - Syndicated loan as a substitute for a bunch of single lender loans
    - Larger amounts than with traditional loans possible
    - Can raise money faster and at better price or non-price terms
- Privacy argument:
    - More flexible terms and conditions

    - Renegotiations possible
    - No information has to be provided to the public about company details (in contrast to the bond prospectus

Leader's View

- Financial perspective:
    - Regulatory or economic limits
    - Diversification (concentration risk)
    - Income / Fees
- Relationship perspective:
    - Enhanced development of other business with the borrower

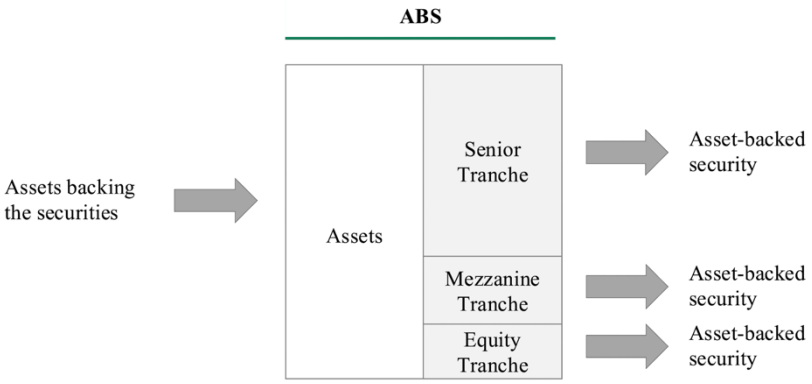
Participant's View

- Diversification (reduce concentration risk)
    - In particular relevant for smaller banks
- Capacity utilization
- Relationship argument (establish new client relationships)

Securitization

ABS=Asset backed securities

- After origination loans of similar characteristics are combined into a portfolio
- Claims are created against the cash flows of this portfolio
- These claims are sold to investors



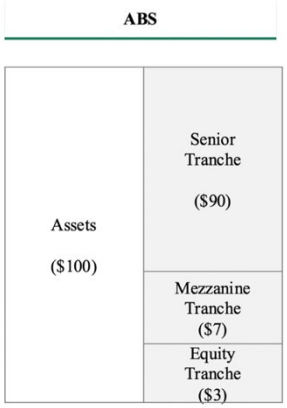
<u>Asset side of an ABS - What are typical assets?</u>

- Residential mortgage loans (RMBS: Residential Mortgaged Backed Securities)
- Commercial real-estate loans (CMBS: Commerical Mortgage Backed Securities)
- Loans (CLO: Collateralized Loan Obligation)
- Bonds (CBO or CDO: Collateralized Bond/Debt Obligation)
- Consumer loans, credit cards (ABS in the narrow sense)
- -> Just examples, no need to learn by heart!

<u>Liability Side of an ABS</u>

- Liability side is tranched (usually 3-6 tranches per deal)
    - Senior tranche has a first claim on the assets, therefore usually AAA rated, but low interest rate
    - Mezzanine tranche with second claim on the assets, higher interest rate, higher default risk

- Equity tranche is residual claimant, highest risk, highest interest rate



Thickness, attachment and detachment points

- Attachement point (AP): Sum of all tranches junior to a tranche
- Detachment point (DP): 100% minus sum of all claimssenior to a tranche
- Example: Mezzanine tranche
  - AP of 3%
  - DP of 10%
  - Thickness: 7%

Valuation

Key Parameters

1. Structure of ABS: Detachment/Atachment Points
2. Interest earned on portfolio of assets
3. Expected loss of portfolio of assets
   - reflects profitability and credit quality of underlying portfolio
4. Correlation of assets in portfolio
   - reflects profitability and credit quality of underlying portfolio
   - Low correlation: higher likelihood of losses
   - High correlation: Higher likelihood of very low or very high losses

Correlation and valuation of tranches

- Low correlation:
  - Lower value for equity tranche
  - Higher value for senior tranche
- High correlation
  - Higher value for equity tranche
  - Lower value for senior tranche

Rationale for securitization: Originator

- Risk management
  - credit risk
  - interest rate risk
  - liquidity risk
- Focus on brokerage activities
  - Origination
  - credit assessment
  - servicing
- Lower capital requirements
  - RWA decrease

Limits to securitization

- Private information / Adverse selection
  - originator knows more about the underlying assets than investors
- Moral hazard
  - originator may put less effort into credit assessment and monitoring

- Outcome
    - Banks sued by investors for not properly screening borrowers before packaging and selling securitizations
    - Regulators put up rules that require a) „skin-in-the-game“ (originator has to keep a certain portion of the ABS) and b) proper due diligence

Financing of M&A transactions
Case Study: Bayer/Monsanto and Amazon/Whole Foods

Questions 5

<u>Are short maturities a substitute for covenants/performance pricing?</u>
Yes…
- Because short maturities mean that loans need to be renegotiated very frequently, so interest rate adjusts to new creditworthiness (e.g., rating)

No…
- Might be much more sensitive than performance pricing
- Includes funding/liquidity risk (loans might not only get more expensive, but loans might not be available at all to the borrower)

In summary: Short maturities are an imperfect substitute to performance pricing and covenants. Usually used in areas where performance pricing or covenants is hard to implement (interbank, sovereign, cross-border).

<u>Securitization & Financial Stability</u>
Reasons why Securitization can enhance financial stability…
- <u>Liquidity risk:</u> Lower risk of a bank run if long-term loans are financed via securitization. More generally: less liquidity risk for banks.
- <u>Interest rate risk:</u> With securitization, assets are financed without maturity mismatch. Thus, bank profits are less sensitive to interest rate changes.
- <u>Credit risk:</u> Credit risk is borne by securitization investors. As long as these are better able to absorb losses (e.g., wealthy or long-term investors), securitization may enhance financial stability.
    - In short (Liquidity/IR/Credit risk): banks not involved in QAT anymore, so securitization helps to reduce structural mismatch on bank balance sheet
- <u>Diversification of risks:</u> risks are spread to many investors, implying that each only bears a small portion of the overall risk
    - Similar: Splitting of medium risk assets into high risk and low risk tranches. If high risk tranches end up with investors that can bear losses, this can enhance financial stability

Reasons why securitization can reduce financial stability …
- <u>Private information / adverse selection</u>: originator knows more about the underlying assets than the investor. With securitization, there is therefore a smaller incentive to screen borrowers.
- <u>Moral hazard:</u> originator may put less effort into credit assessment and monitoring.
- <u>Complexity/Opaqueness:</u> losses are spread to many investors who might not be able to bear the risk (e.g. institutional investors, money market funds, etc) and which might increase systemic risk in the banking sector.

Lecture 6 – Bank Funding

Deposit Contract

- Debt contract, non-traded
- Available on demand
- Sequential service constraint
- Two Key Rationales: 1. Depositors are insured against liquidity shocks & 2. Depositors can discipline banks

Liquidity Insurance (Diamond & Dybvig Model)

Result 1: Household behavior without financial intermediation

- In t=0 households invest in the project rather than storing
- This implies that 'early' Consumers will liquiditate projects

Result 2: Demand Deposits as Liquidity Insurance (with financial intermediation)

- A bank can prevent costly liquidation of projects by pooling liquidity risk across borrowers
- A bank can provide additional consumption smoothing by pooling resources and reallocating them between late and early consumers
- Note: Welfare is higher with financial intermediation even if the level of investment in the economy is lower than without banks

Bank Runs

- Payoff to any specific consumer depends on withdrawal behavior of other consumers
- There is potential for a bank run due to „pure panic"

Result 3: no panic equilibrium

- The withdrawal by all late consumers in t=2 is a Nash Equilibrium
- Given that all other late consumers do not withdraw in t=1 it is optimal for any particular household "X" not to withdraw in t=1

Result 4: "panic" bank run equilibrium

- The withdrawal by all late consumers in t=1 is also a Nash Equilibrium

Deposit Insurance

- Implication: Lower incentive to run

Result 5: deposit insurance

- The presence of a deposit insurance scheme makes a "panic" driven bank-run less likely

Result 6: no sequential service constraint

- In the absence of a sequential service constraint a "panic" driven bank-run is less likely

Market Discipline

- With on-demand deposits and a sequential service constraint, depositors have incentives to discipline banks (withdraw their money when the bank's creditworthiness declines)
- This disciplines banks ex-ante and avoids that bank take on too much risk
- The absence of a sequential service constraint or the existence of deposit insurance diminish the disciplining incentives for informed depositors
    - Market discipline more likely with unsecured short-term funding such as funding from money market mutual funds or interbank funding

Theories of Bank Capital Structure

- Modigliani-Miller: The market value of a firm (sum of the market values of debt and equity) is independent of its capital structure
    - Assumptions: Perfect capital markets, no transaction costs, etc.
    - Prediction: Bank value should be unrelated to bank leverage
- Trade-off theory: Optimal capital structure balances tax-shields against costs of financial distress
    - Banks have low asset risk because their assets consist of liquidity and loans
    - Prediction: Banks should have a much higher leverage than non-banks
- Liquidity Production: Deposits are an essential part of financial services business model, not just a source of capital
    - Deposits provide liquidity on-demand, and can therefore be raised at lower costs
    - Prediction: Banks should have a high leverage
- Market Discipline: Sequential servicing constraint of deposits (or generally: of short-term liabilities) leads to disciplining of the bank
    - Threat of withdrawal disciplines banks
    - This discipline implies that deposits are risk-free (any risky behavior of banks would lead to a bank-run, and thus banks do not engage in too risky activities in the first place)
    - Prediction: Banks should have a high leverage
- Government safety net: In contrast to non-financial firms, banks benefit from a significant government safety net
    - Deposit insurance
    - Access to central bank funding
    - Too-big-to-fail / bank bail-outs in crises
    - Prediction: Banks have a high leverage (at the expense of tax payers' money)

Lecture 7 – Financial Crisis

IMF Definition

1) Significant signs of financial distress in the banking system (as indicated by significant bank runs, losses in the banking system, and/or bank liquidations).

i. a country's banking system exhibits significant losses resulting in a share of nonperforming loans above 20% of total loans or bank closures of at least 20% of banking system assets or
ii. fiscal restructuring costs of the banking sector are sufficiently high, exceeding 5% of GDP

AND

2) Significant banking policy intervention measures in response to significant losses in the banking system. Criteria fulfilled if 3/6 measures have been used:

1. Deposit freezes / bank holidays
2. Significant bank nationalizations
3. Bank restructuring fiscal costs (at least 3% of GDP)
4. Extensive liquidity support (at least 5% of deposits and liabilities to nonresidents)
5. Significant guarantees put in place
6. Significant asset purchases (at least 5% of GDP)

Typical Responses (in high income countries)

Financial Sector Interventions

- Recapitalizations
- Extensive liquidity support
- Guarantees on bank liabilities
- Nationalizations
- Asset Purchases
- Measures are less probable if you are in a low/middle income country, because government spending is limited and can endager their debt level

Macro and Fiscal Policy Responses

- Central bank lowers short-term interest rates
- Significant fiscal deficits in the years after the crisis

Typical Costs

- Output loss much higher than Net Fiscal Costs, because output losses also includes indirect effects (lower growth, worse allocation of capital, higher unemployment)

Case Studies / Examples

- 1. United States 1988: Savings and Loans crisis
  - Main weakness exposed: Maturity transformation
  - Resulting changes in bank business models: Adjustable-rate mortgages, Securitization
- 2. Nordic Crisis 1991
  - Main weakness exposed: Risky lending
- 3. Japanese Crisis 1998
  - Main weakness exposed: Real estate lending
- 4. Global financial crisis 2007/2008
  - Main weakness exposed: Securitization, Runs in the shadow banking market

General Learnings

Causes

- Financial crises usually arise from the qualitative asset transformation function of banks (liquidity transformation, maturity transformation, credit risk transformation)
- Often driven by credit booms that collapse
- Real estate is frequently central due to large volumes and long asset lifetimes
- Deregulation can fuel credit booms via rapid, reversible capital inflows
- Bank runs typically amplify rather than cause crises

Policy responses (no universal solution)

- Common recommendation in banking crises is to act quickly and be transparent about losses
- Recapitalize banks, often separating them into good banks / bad banks
- Avoid evergreening (rolling over bad loans to hide losses)

Lecture 8 – Banking Regulation

Reasons for Bank Regulation

Argument 1: Deposit insurance / Government safety net

- Deposit-taking is welfare-enhancing, but deposit-taking institutions are prone to runs
- Deposit insurance and access to central bank funding leads to moral hazard of banks
    - incentive to increase leverage, provide risky loans, decrease liquid assets
- Externalities on tax-payers
- Solution: Requirement to have sufficient capital (to absorb losses from lending) and liquidity (to cover withdrawals)

Argument 2: Too-big-to-fail

- Failure of large banks can have externalities on financial system and real economy
    - declines in trust in other financial intermediaries, leading to bank runs
    - bank's customer: hard to substitute loss of funding (because of lending/information asymmetries)
- Solution: Requirement to have sufficient capital (to absorb losses from lending) and liquidity (to cover withdrawals)

Argument 3: Systemic risk

- Failure of many large banks can have significant externalities on financial sector and real economy
    - declines in trust in other financial intermediaries, leading to bank runs
    - bank's customer: hard to substitute loss of funding (because of lending/information asymmetries)
- Solution: Requirement to have sufficient capital (to absorb losses from lending) and liquidity (to cover withdrawals)

Argument 4: Consumer protection

- Financial products different from traditional products
    - High information asymmetry between bank and customer
    - Hard to assess product quality and no right-of-return
- Solution: rules on transparency of markets and quality of financial advice

Argument 5: Most convenient target to enforce law in other areas

- Recently, non-financial risk has become increasingly important for banks
- Key areas: Fraud, Financial crime and conduct risk

Minimum reserve requirements

- Requirement to hold a certain amount of funds as reserves in their accounts at the central bank
    - Euro Area: 1% of "money-like" liabilities (ECB pays no interest)
    - Has little impact in practice

Capital Regulation

- Leverage restrictions to reduce externalities (Argument 1,2,3) and have a buffer
    - reduces banks' incentives to take risk

Structure of equity requirement

$$Regulatory\ Capital\ \geq Hurdle\ Rate\ \cdot RiskWeightedAssets$$

Regulatory Capital

- resources of a bank which can absorb potential losses
- Tier 1 capital: loss absorbing when bank is a 'going concern'
- Tier 2 capital: Loss absorbing when bank is a 'gone concern'

Hurdle Rate

- Minimum required amount invested to be deemed acceptable
- Basel I: Minimum Tier 1 -> 4% of RWA
- Basel II (Pillar 1): More risk sensitive capital requirements (4% Tier 1 / 8% Tier 1 + Tier 2 remained unchanged)

Risk weighted assets (RWA)

- A bank's loans and other assets weighted according to risk to determine the minimum amount of capital it must hold

Lessons from the 2007-2009 financial crisis

- Capital levels are too low
- Weak market discipline due to opacity of exposure (OTC contracts, securitization)
- Procyclicality due to market-sensitive risk assessments

Basel III

- Stricter capital requirements! (in % risk weighted assets)
    - Capital requirements depend on bank size and business cycle
- Leverage ratio! -> Minimum capital ratio: 3%
- Stricter assessment of on-balance / off-balance risks
- Liquidity regulation

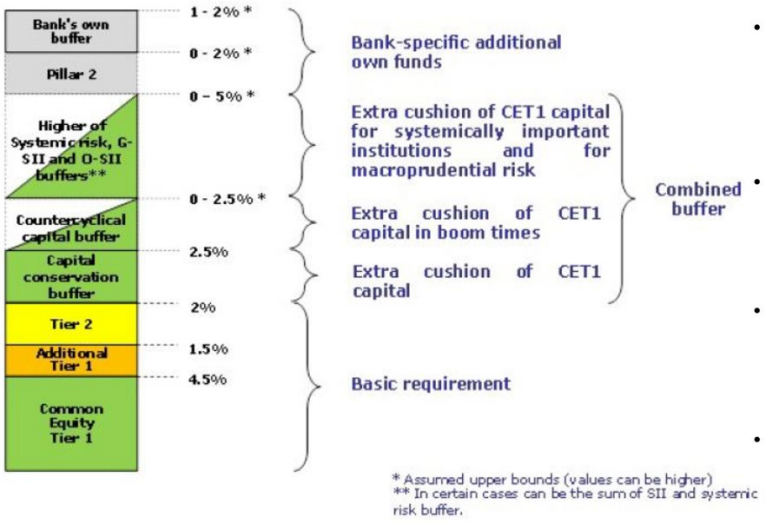
Basel III Capital requirements

Capital conversation buffer: when a bank breaches the buffer, automatic safeguards apply to limit the amount of dividend and bonus payments it can make

Countercyclical capital buffer: high in good times, low in bad times -> designed to help counter pro-cyclicality in the financial system

Systemic risk buffers: address the potential negative effects that systemic institutions may have on the international or domestic financial system

Bank-specific add-ons: at the discretion of the regulator (ECB, national regulators)



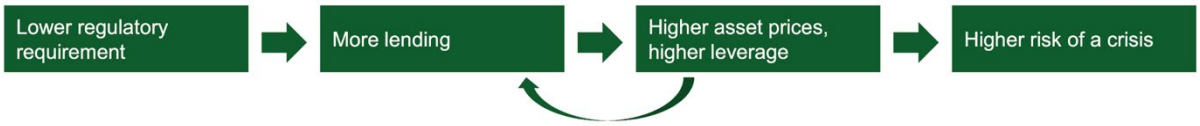
Shall capital requirements be risk-weighted?

Arguments for risk-weighted capital requirements

- Buffer function of capital: Banks with less risky assets require a smaller buffer
- Incentive function of capital: if capital requirements are independent of risk-levels, banks have incentives to take more risk

Arguments for a simple unweighted leverage ratio

- Misreporting of risk weights: Studies have shown that poor-capitalized banks misreport / manipulate their risk weights. Leverage ratio can be less prone to manipulation.
- Endogenous risk: Low risk weights can encourage lending, thereby fuelling a credit boom, which later turns into a bust
    - Endogenous risk = created by interaction of players in the market and regulations
    - Typical manifestation of endogenous risk in the banking sector:
        - Low risk weights encourage loans in a particular sector
        - An increase in loan volume pushes up asset prices and increases leverage of the borrowers
        - Higher asset prices further encourage lending (more collateral, lower historical default rates)
        - High asset prices and high leverage increase likelihood of a severe crisis



Basel IV / Finalisation of Basel III

- Main part: Output floor of 72.5%
    - Banks' calculations of RWAs generated by internal models cannot, in aggregate, fall below 72.5% of the risk-weighted assets computed by the standardised approaches

Liquidity Regulation

Liquidity coverage ratio (LCR)

- Regulation of short-term liquidity
- Formal requirement:

$$\frac{Stock\ of\ HQLA}{Total\ net\ cash\ outflows\ over\ the\ next\ 30\ calendar\ days} \geq 100\%$$

- HQLA = High-quality liquid assets
    - General requirement: should be liquid in markets during time of distress
    - Level 1 and Level 2
- Net cash outflows: based on actual behavior, not legal maturity. Key area of discussion: outflow assumption for deposits (ranges from 3% for deposits to 40% for interbank loans)

Net stable funding ratio (NSFR)

- Regulation of structural, longer-term funding mismatch

- Formal requirement:

$$\frac{Available\ amount\ of\ stable\ funding}{Required\ amount\ of\ stable\ funding} \geq 100\%$$

- ASF (Available stable funding) and RSF (required stable funding – long- term assets) factors are used to determine available and required amount of stable funding

Lecture 9 – Money

Money – The Basics

Meaning of Money

- anything that is generally accepted in payment for goods or services or in the repayment of debts

Functions of Money (BCS)

1. Medium of Exchange

- Eliminates the trouble of finding a double coincidence of needs (reduces transaction costs)
- Promotes specialization
- A medium of exchange must:
    - be easily standardized
    - be divisible
    - be easy to carry
    - not deteriorate quickly
    - be widely accepted (=social convention)

2. Unit of Account

- used to measure value in the economy
- reduces transaction costs

3. Store of Value

- used to save purchasing power over time
- other assets also serve this function but Money is the most liquid of all assets

Beyond the three main functions

4. Transaction efficiency:

5. allowing transactions to be conducted in a reliable, timely, and cost-efficient manner

5. Financial inclusion:

- allowing broad access to money
- Note: 4./5. can be seen as a sub-function of "Medium of Exchange"

6. Privacy:

- protecting individuals' privacy

7. AML (Anti-Money Laundering)/CFT (Countering Financing of Terrorism):

- prevent use for illegal purposes

8. Efficiency and stability of the financial system:

- mitigate the prospect of disruptions in the financial intermediation process that are severe enough to adversely impact real economic activity

Evolution of money

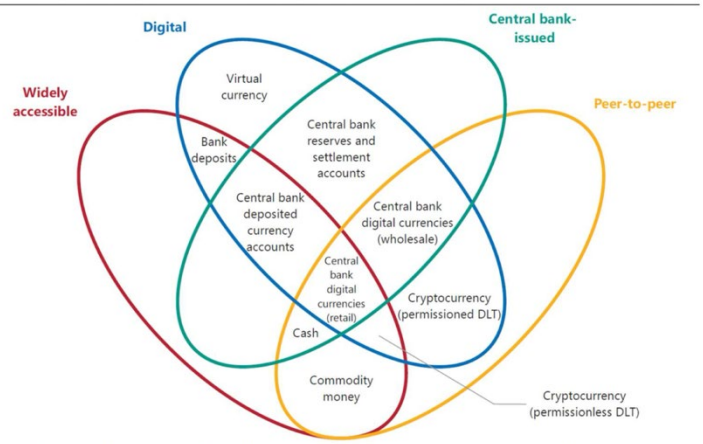
- Commodity Money: valuable, easily standardized and divisible commodities (e.g. precious metals, cigarettes)
- Fiat Money: paper money decreed by governments as legal tender

Claims on fiat money

- Checks: an instruction to your bank to transfer money from your account
- Bank deposits: Digital claim on fiat money

Taxonomy of Money

The money flower: a taxonomy of money — Graph V.1



Source: Adapted from M Bech and R Garratt, "Central bank cryptocurrencies", *BIS Quarterly Review*, September 2017, pp 55–70.

Widely accessible:
- Necessary feature to become a general medium of exchange
- Operationally more demanding than money that is only accessible by few partcipants

Digital:
- Facilitates efficient payment
- Can compromise privacy; requires access to digital infrastructure

Central bank-issued:
- Backed by governmant authority
- Requires trust in government

Peer-to-Peer:
- No intermediary necessary
- If in digital form, typically requires technical skills to access

Monetary aggregates: Measuring money supply

Construct monetary aggregates using the concept of liquidity:
- M1 (most liquid assets) = currency + overnight deposits
- M2 (adds to M1 other assets that are not so liquid)
    - M1 + other short-term time deposits
- M3 (adds to M2 other assets that are not so liquid)
    - = M2 + marketable instruments (Repos, Money Market Fund shares, short-term debt securities issued by banks)

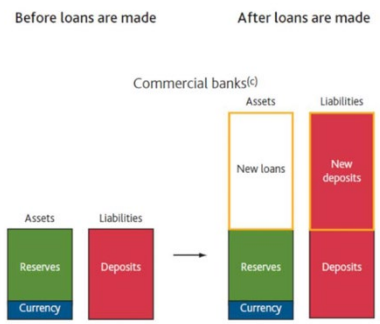
⇒ Overnight deposits are by far the most important component of monetary aggregates
⇒ Intimately links lending and deposit-taking, and facilitates transformation functions
⇒ Increase in monetary aggregates over past decades

Money Creation in Modern Economies

Lending creates deposits
- Banks create money by making loans
- The banking system does NOT lend out deposits that savers place with them

- The opposite is true: the banking system creates deposits out of thin air when making a loan
- Creating deposits out of thin air ≠ banking sector can print money for themselves
- A deposit is a liability for banks, not an asset
- Three main sets of constraints that restrict the amount of deposits that banks can create
    1. Limit on how much banks can lend (profitability of lending, riskiness of lending, regulatory policies)
    2. Constraints by holders of deposits (households, businesses): For example, household might decide to use deposits to repay loan
    3. Monetary policy: influences interest rates, and thus the amont that households and businesses want to borrow

Figure 1 Money creation by the aggregate banking sector making additional loans[a]



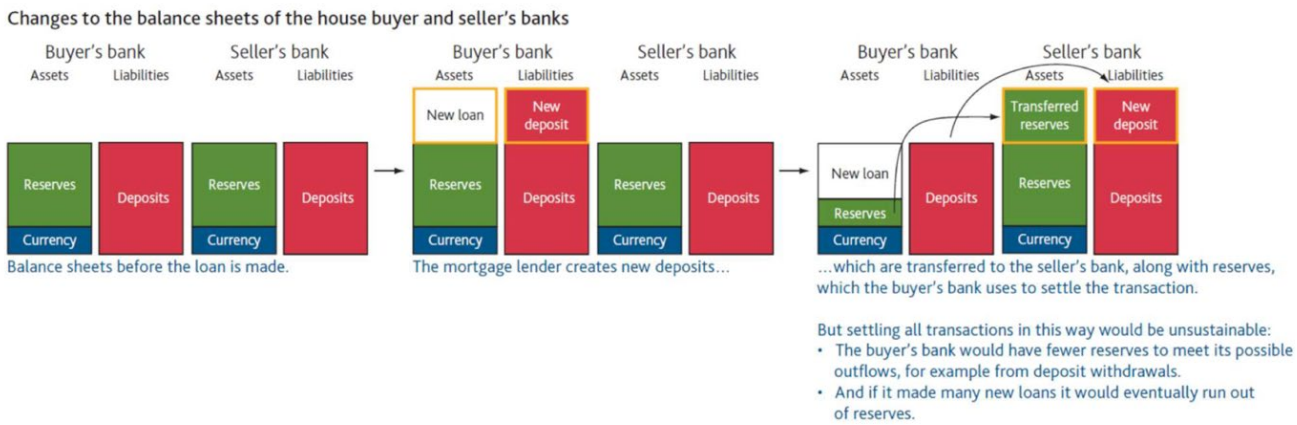
<u>Individual banks: Deposits needed to grant loans</u>

- When an individual bank makes a loan, it simultaneously creates a deposit out of thin air
- But: this deposit is likely going to be used to pay for something, which implies that the deposit is transferred to a different bank
- The bank thus needs to retain new deposits to accompany the new loan

Side Notes

- Reserves are used only for interbank settlement, not for lending per se
- Settlement is usually netted -> Banks don't transfer reserves for *every* transaction
    - Payments are offset during the day and settled net at the end (RTGS systems, clearing houses)

Figure 2 Money creation for an individual bank making an additional loan[a]



Payments

Takeaway: Payments

- Existing infrastructure provided by Eurosystems: Real time, Cheap, Secure and reliable
- Existing infrastructure typically not available and/or not sufficient for: for cross-border transactions, E-Commerce, and illicit payments

Lecture 10 – Special Topics: Stablecoins

Definition and overview

- A crypto-asset that aims to maintain a stable value relative to a specific asset, or a pool of assets
- Mechanisms to maintain a stable value
    - Reserve assets (by far the dominant form of stablecoins)
    - Algorithmic (rarely used, not discussed further here)
- Typically pegged to a currency, in particular the USD
- Wide use in the cryptosphere, in particular to trade crypto-assets.
- Some use in cross-border payments, very limited uses beyond.

Risks of stablecoins

- runs on stablecoins
- negative effect on lending
- financial repression
- spillovers to safe asset markets

Key advantages compared to traditional payment rails

- Can enable programmable payment: but: uses cases limited to quite special cases.
- Regulatory arbitrage (AML/CFT), but: not dependent on technology, but will of regulators